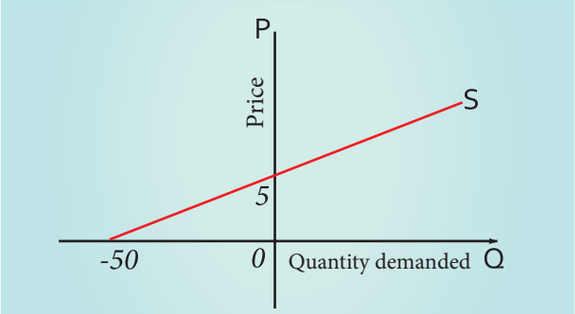


Diagram 12.4

Then $$Y-6=\frac{X-10}{10}$$

$$10(Y-6)=X-10$$

$$10Y-60=X-10$$

$$10Y-60+10=X$$

$$10Y-50=X$$

$$-X=-10Y+50$$

Multiplying both sides by minus (–), we get

$$X=-50+10Y$$

Considering X as quantity supplied and Y as price (P)

Then X = 10P – 50 (or)

$$\boxed{X=-50+10P}$$

If Price = 0; Q = –50

If Q = 0; P = 5

Note: The coefficient of 'P' is – in demand function.

The coefficient of 'P' is + in supply function.

12.2.4 Equilibrium

The point of intersection of demand line and supply line is known as equilibrium. The point of equilibrium is obtained by using the method of solving a set of equations. One can obtain the values of two unknowns with two equations. At equilibrium point,

Demand = Supply

(These are hypothetical examples)

$$100-10P=50+10P$$

$$100-50=20P$$

$$50=20P$$

$$\frac{50}{20}=P$$

$$P=2.5$$

When P = 2.5, Demand = 100–10 (2.5)

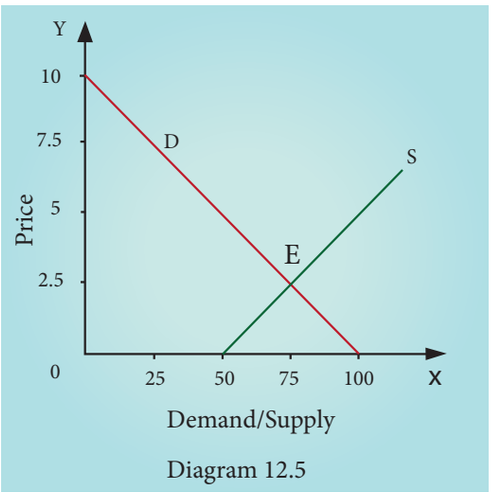
= 75

When P = 2.5, Supply = 50 +10 (2.5)

= 75

Example: 12.3

Find the equilibrium price and quantity by using the following demand and supply functions Qd = 100–5P and Q_s = 5P respectively.



Demand/Supply

Diagram 12.5

Solution:

Equilibrium is attained when,

$$Q_s=Q_d$$

$$5P=100-5P$$

$$10P=100$$

$$P=10$$

When P = 10

In supply function

$$Q_s=5P=5\times 10=50$$

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